Introduction to Algorithms and Data Structures

week 4: Problem Solving

SAXION UNIVERSITY OF APPLIED SCIENCES

- **Discuss work last lesson**
- Problem solving
- Workshop/homework

- Where you able to do the exercises?
  1. Yes! All exercises are done!
  2. No.., Because I have these questions
  3. I haven't prioritized this subject enough at the moment to do the assignments

- **Discuss work last lesson**
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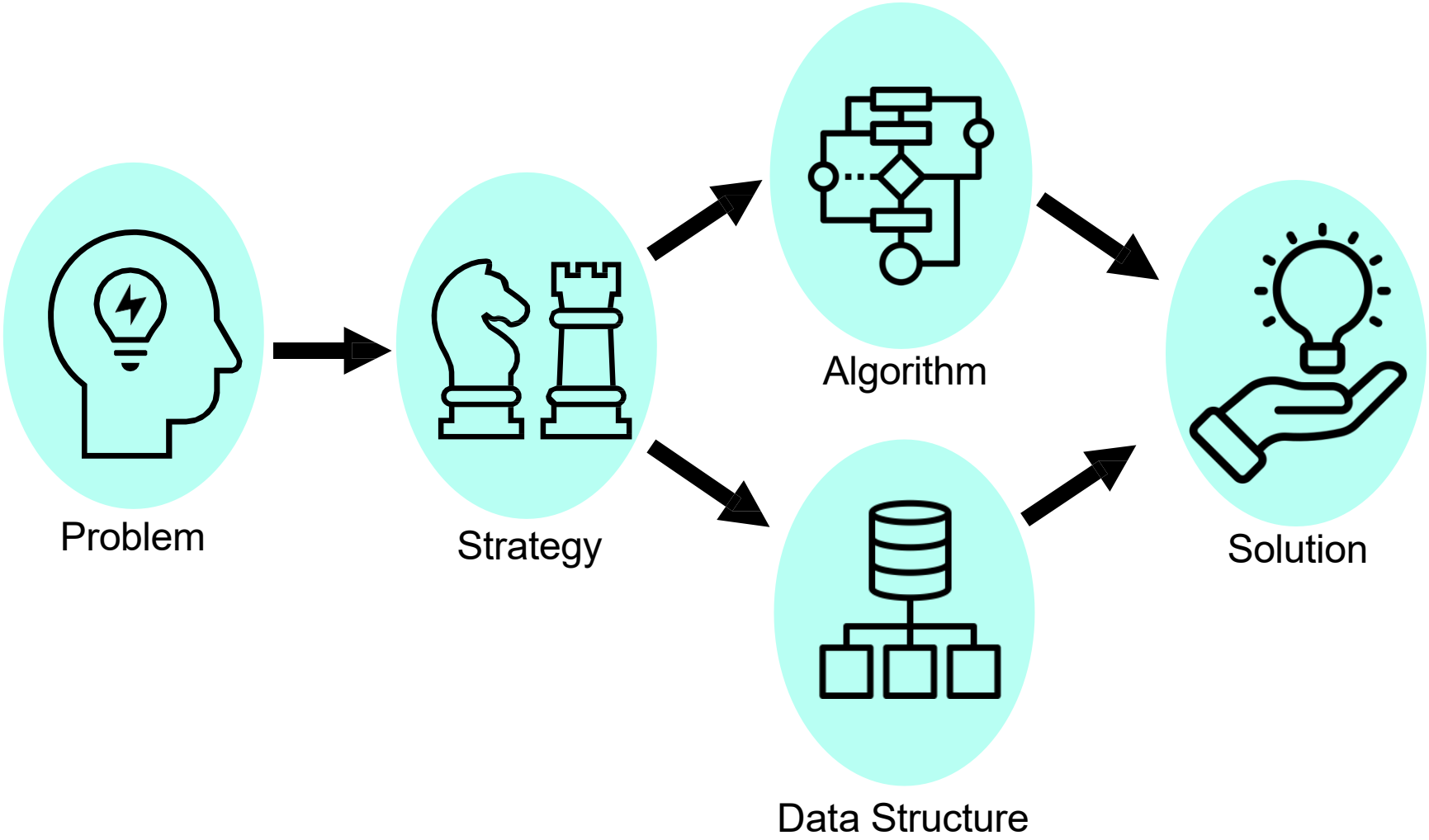
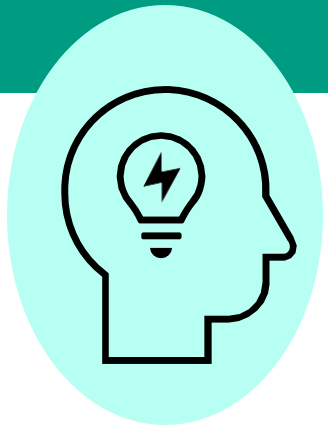
Problem Solving – General approach

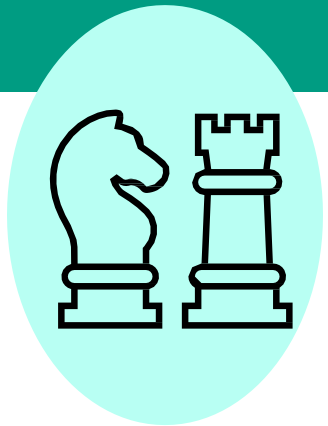


Understanding the problem



- Make sure you fully understand the problem.
- **Read the full description**, ask questions if you can.
- Think about inputs and outputs.
- Read the examples or create some simple ones of your own
  - Create more complex examples / edge cases

Devise a strategy for solving the problem

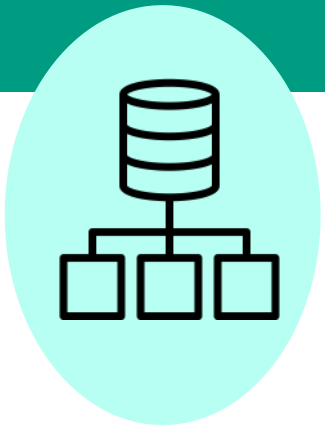


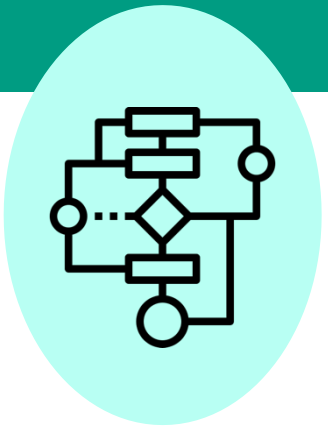
- Try to solve the examples yourself
  - Start with the simple examples
- Divide the problem in smaller steps (divide and conquer / recursion)
  - Try to simplify/shorten the problem until you see a solution.
  - **Don't try to solve a problem that is too big**;

    if you can't solve the problem (yet!),

    then you haven't broken it down into steps that are simple enough!

Devise an algorithm with (a) data structure(s)





- Code your previous steps in pseudocode (your algorithm) and think of applicable data structures to use.

Solve the problem

- Code your pseudocode; translate pseudocode to Java

  (or any other programming language you are working with).
- Test often (fail fast);
  - don't code your whole solution in one run, because finding bugs will be much harder, so
  - **code and immediately test small steps**.
- **Use the debugger** to see what's happening.
- Start with the easiest solution you can think of;
  - Working code trumps optimized failure.
  - Optimization comes later, and only if you really need to!

This week: Problem solving

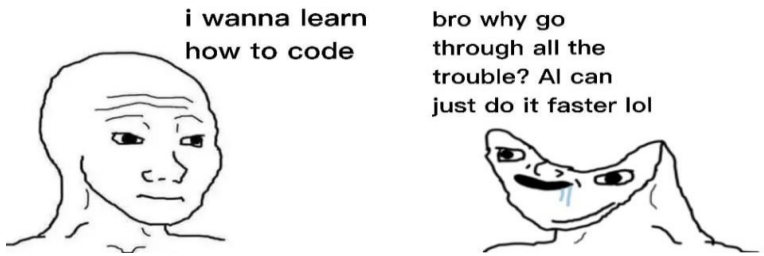
- No new concepts!
- But, challenges become larger,
- and more difficult.
- Time usage:

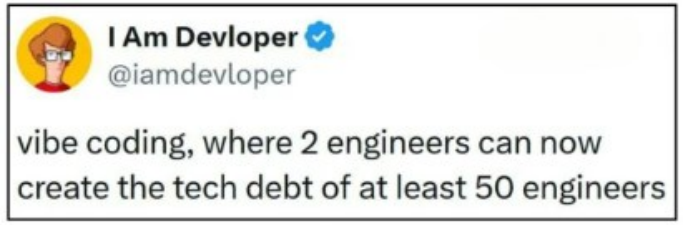
programming << thinking

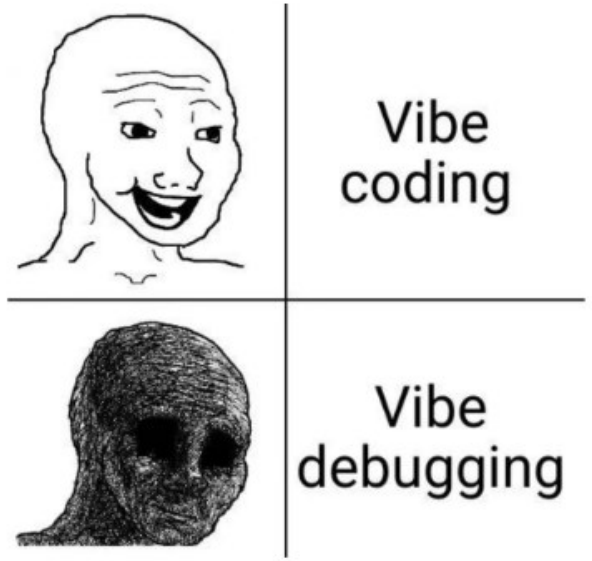
Using AI?

- Use an AI like ChatGPT and be done with the exercises in 5 minutes
  - Pro: saves some time
  - Downside: you don't learn problem-solving
  - Downside: you don't really understand the code
  - Downside: you don't learn how to find & fix bugs

exam failed!







- Instruct an AI to act as a tutor to explain concepts, and you have a 24/7 instructor.

- Discuss work last lesson
- What is recursion
- Workshop/homework

Now it’s your turn!